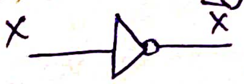
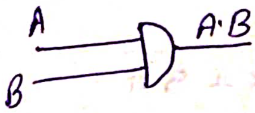
DIGITAL ELECTRONICS

NOT GATE (uniary operator)



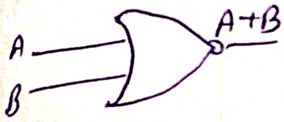
A	$\bar{A}$
0	1
1	0

AND GATE (binary operator)



A	B	Y=A·B
0	0	0
0	1	0
1	0	0
1	1	1

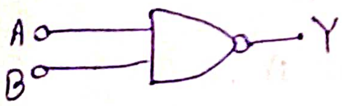
OR GATE



A	B	Y= A+B
0	1	1
0	0	0
1	1	1
1	0	1

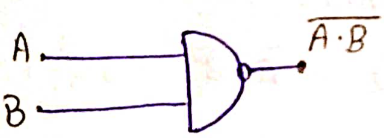
UNIVERSAL GATES.

(1). NOR GATE



A	B	$Y=\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

(2). NAND GATE

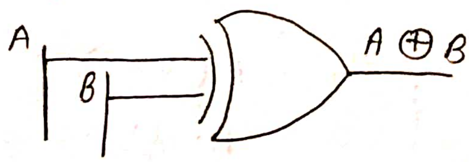


A	B	$Y=\overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

(3). XOR GATE (Exclusive or Gate). [XOR Gate produces output 1 for only those input combination that have odd no. of ones.

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

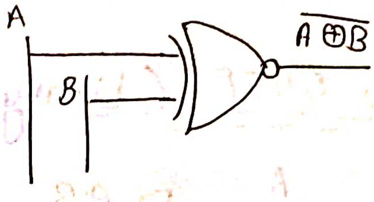
$$A \oplus B = A \cdot \bar{B} + B \cdot \bar{A}$$



(4). XNOR GATE (NOT XOR GATE)

It produces output 0 for only those, whose input combination have odd no. of ones.

A	B	$\overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1



$$Y = \overline{A\cdot\bar{B} + B\cdot\bar{A}}$$

$Y =$

★ Some Imp. Laws.

(1). $A + 0 = A$

(2). $A \cdot 0 = 0$

(3). $A + 1 = 1$

(4). $1 \cdot A = 1$

(5). $A + A = A$

(6). $A \cdot A = A$

(7). $A + \bar{A} = 1$

(8). $A \cdot \bar{A} = 0$

(9). $\bar{\bar{A}} = A$

(10). $A + B = B + A$

(11). $A \cdot B = B \cdot A$

(12). $A(B+C) = (A+B)+C$

(13). $A \cdot (B \cdot C) = (A \cdot B) \cdot C$

(14). $A(B+C) = A \cdot B + A \cdot C$

(15). $A + (B\,C) = (A+B)(A+C)$

(16). $A + AB = A$.

(17). $A \cdot (A+B) = A$.

(18). $A + \bar{A}B = A + B$.

(19). DEMORGAN'S LAW.

(i). $\overline{(A+B)} = \bar{A} \cdot \bar{B}$

(ii). $\overline{A \cdot B} = \bar{A} + \bar{B}$

$$xy + yz + \bar{x}z = xy + \bar{x}z$$

LHS $= xy + 1 \cdot yz + \bar{x}z$

$xy + (x+\bar{x})yz + \bar{x}z$

$xy + xyz + \bar{x}yz + \bar{x}z$

$xy(1+z) + \bar{x}z(y+1) = xy + \bar{x}z$

* Boolean expression can be of 2 types :-
(1). Minterm (SOP) :- A minterm is a product of all the literals with in the logic system.

* CANONICAL EXPRESSION
A boolean expression composed entirely either of minterm or maxterm, is reffered to as canonical expression.

Q. Write S.O.P. :-

x	y	z	F	S.O.P
0	0	0	0	
0	0	1	0	
0	1	0	1 ←	$\bar{x}\cdot y\cdot\bar{z}$
0	1	1	1 ←	$\bar{x}\cdot y\cdot z$
1	0	0	0	
1	0	1	0	
1	1	0	0	
1	1	1	1 ←	xyz

$$S.O.P = f = \bar{x}y\cdot\bar{z} + \bar{x}yz + xyz$$

Q. Write S.O.P

16 8 4 2 1

$$F(x,y,z) = \Sigma(0,3,4)$$

$$m_0 = 000 = \bar{x}\bar{y}\bar{z}$$

$$m_3 = 011 = \bar{x}yz$$

$$m_4 = 100 = x\bar{y}\bar{z}$$

$$S.O.P = \bar{x}\bar{y}\bar{z} + \bar{x}yz + x\bar{y}\bar{z}$$

(2). MAXTERM (P.O.S) :- A maxterm is product of sum of all the literal with in the logic system.

Q. Write POS :-

$$Y = \bar{A}\cdot(B+C)$$

$$Y = (\bar{A} + B\bar{B} + C\bar{C})\ (A\bar{A} + B + C)$$

$$= (\bar{A} + B\bar{B} + C)\cdot(\bar{A} + B\bar{B} + \bar{C})\cdot(A+B+C)(\bar{A}+B+C)$$

$$= (\bar{A} + B\bar{B} + C)\cdot(\bar{A} + \bar{C} + B)(\bar{A}+\bar{B}+\bar{C})\cdot(A+B+C)$$

$$= \cancel{(\bar{A}+B+C)}\ (\bar{A}+\bar{B}+C)\cdot(\bar{A}+\bar{C}+B)(\bar{A}+\bar{B}+\bar{C})\cdot(A+B+C)(\bar{A}+B+C)$$

$$= (A+B+C)(\bar{A}+B+C)\cdot(\bar{A}+\bar{B}+C)\cdot(\bar{A}+B+\bar{C})\cdot(\bar{A}+\bar{B}+\bar{C})$$

(ii). $(x+y)(y+z)(z+x)$

$(x+y+z\bar{z})(x\bar{x}+y+z)(x+y\bar{y}+z)$

$= (x+y+z)(x+y+\bar{z})\cancel{(x+y+z)}(\bar{x}+y+z)\cancel{(x+y+z)}(x+\bar{y}+z)$

$= (x+y+z)(\bar{x}+y+z)(x+\bar{y}+z)(x+y+\bar{z})$

Q. Write P.O.S :-

x	y	z	F	POS
0	0	0	1	
0	0	1	0 ←	$x+y+\bar{z}$
0	1	0	1	
0	1	1	0 ←	$x+\bar{y}+\bar{z}$
1	0	0	1	
1	0	1	0 ←	$\bar{x}+y+\bar{z}$
1	1	0	1	
1	1	1	1	

$P.O.S = (x+y+\bar{z})(x+\bar{y}+\bar{z})(\bar{x}+y+\bar{z})$

Q. Write P.O.S :-

$F(a,b,c,d) = \prod(1, 13, 14, 15)$

$m_1 = 0001 \rightarrow A+B+C+\bar{D}$

$m_{13} = 1101 \rightarrow \bar{A}+\bar{B}+C+\bar{D}$

$m_{14} = 1110 \rightarrow \bar{A}+\bar{B}+\bar{C}+D$

$m_{15} = 1111 \rightarrow \bar{A}+\bar{B}+\bar{C}+\bar{D}$

$POS = (A+B+C+\bar{D})(\bar{A}+\bar{B}+C+\bar{D})(\bar{A}+\bar{B}+\bar{C}+D)$
$(\bar{A}+\bar{B}+\bar{C}+\bar{D})$.

KARNAUGH MAP (K-MAP).

(i). FOR S.O.P

For (2).

A \ B	$\bar{B}$	B
$\bar{A}$	0	1
A	2	3

For 3 variables.

A \ BC	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	1	3	2
A	4	5	7	6

For 4 var.

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	1	3	2
$\bar{A}B$	4	5	7	6
AB	12	13	15	14
$A\bar{B}$	8	9	10	11

Q. $F(a,b,c,d) = \Sigma(0,2,3,4,6,7,8,10,12)$

AB \ CD	$\bar{C}\bar{D}$ (I)	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1 (0)	(1)	1 (3)	1 (2)
$\bar{A}B$	1 (4)	(5)	1 (7)	1 (6)
AB	1 (12)	(13)	(15)	(14)
$A\bar{B}$	1 (8)	(9)	(11)	1 (10)

(II)

(III).

$$S.O.P = \bar{C}\bar{D} + \bar{A}C + \bar{B}\bar{D}$$

OR.

I^{st} Quad $= m_0 + m_4 + m_{12} + m_8$

$= \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + AB\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D}$

$= \bar{C}\bar{D}$ (after reducing AB)

II^{nd} Quad $= m_3 + m_2 + m_7 + m_6$

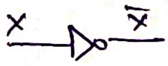
$= \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}BCD + \bar{A}BC\bar{D}$

$= \bar{A}C$

III^{rd} Quad. $= m_0 + m_2 + m_8 + m_{10} = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D}$

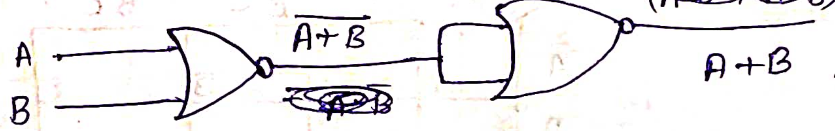
$= \bar{B}\bar{D}$.

BASIC GATES USING NOR GATES.

(1). NOT → 

(2). ~~AND~~ OR 

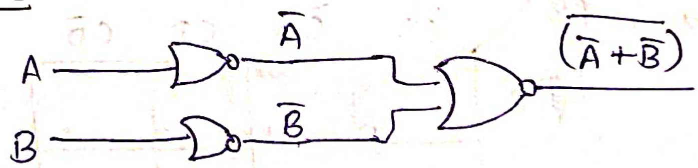
$\overline{(\overline{A}\cdot\overline{B}+\overline{A}\cdot\overline{B})} = \overline{(\overline{A}\cdot\overline{B})} = A\cdot B$



$A+B$.

(3). ~~OR~~ 

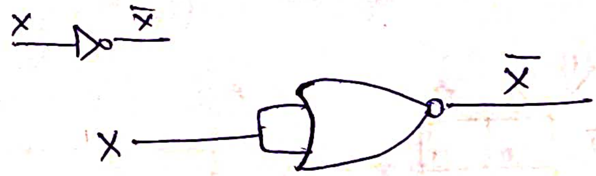
AND



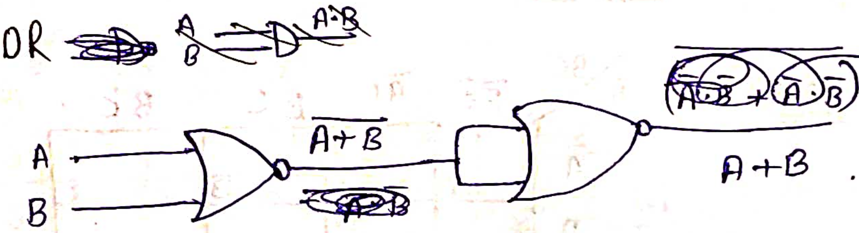
$\overline{(\overline{A}+\overline{B})} = \overline{\overline{A}}\cdot\overline{\overline{B}} = A\cdot B$

BASIC GATES USING NOR GATES.

(1). NOT →

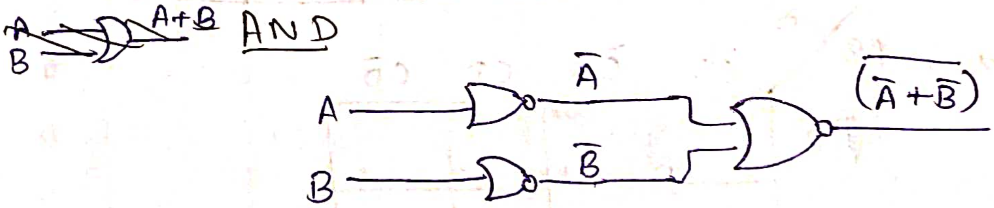


(2). ~~AND~~ OR



$\overline{\overline{A+B}} \Rightarrow A\cdot B$

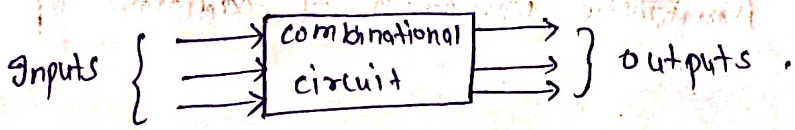
(3). ~~OR~~ AND



$\overline{(\bar{A}+\bar{B})} = \bar{\bar{A}}\cdot\bar{\bar{B}} = A\cdot B$

COMBINATIONAL CIRCUITS:

The output of combinational circuit depends on its present inputs only.



- ★ Combinational circuits perform a specific information processing operation Fully specified logically by a set of Boolean functions.
- ★ Combinational consists of input variable, output variables & logic gates.
- ★ Logic gates accept signals from the inputs & generate signals to the outputs.

DESIGN PROCEDURE.

(1). No. of available input variable & req. output variables are determined.

(2). The input & output variable are assigned letter symbols.

(3). Truth table is derived that defines the required relationship b/w inputs & outputs variables.

(4). Simplified Boolean function for each output is obtained.

(5). Then, the logic diagram is draw with the help of logic gates.

- ★ Memory unit is not required.
- ★ It is faster than seq. circuits becoz delay is only due to propogation delay of gates.
- ★ It is easy to design.

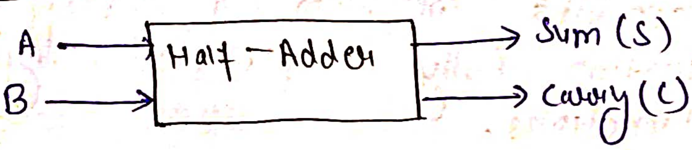
Ex→ Adder, Code Converter, multiplexer, Subtractor, etc.

HALF ADDER

It adds the 2 inputs (A & B) & produces the sum (S) & the carry (C).

It perform the Arithmetic operation of addition of 2 single bit words.

Block Diagram :-



Truth Table.

Input		Output	
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$$\begin{array}{r} 1 \\ +1 \\ \hline 2 \end{array} \rightarrow \text{Binary} = 10$$

K-Map for S & C.

for S →

B \ A	0	1
0	$_0$	$\boxed{1}_1$
1	$\boxed{1}_2$	$_3$

$S(A,B) = \Sigma(1,2)$

→ XOR Gate.

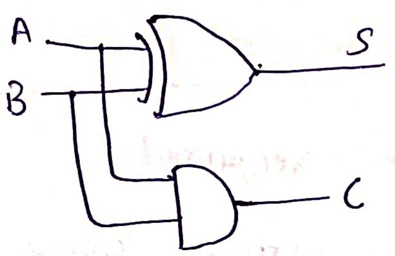
$S = \bar{A}B + A\bar{B} = A \oplus B$

~~Logic Diagram.~~ for C →

B \ A	0	1
0	$_0$	$_1$
1	$_2$	$\boxed{1}_3$

$C = AB$

Logic Diagram.



↓

(Half - Adder)'s logic Diagram.

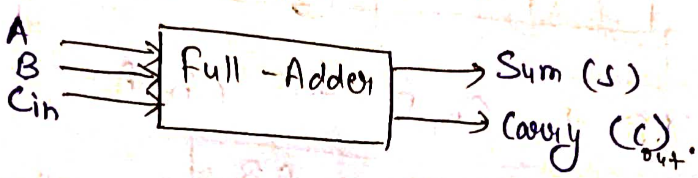
✱ It does not take carry from previous sum.

FULL ADDER

A Full adder is a comb. circuit that adds 2 bits & carry ~~that adds~~ and outputs a sum bit & a carry bit.

It has 3 input & 2 output.

Block Diagram



Truth table

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

S C
0 1

K-MAP for S & Cout

for S.

Cin \ AB	00	01	11	10
0	0	1 1	3	1 2
1	1 4	5	1 7	6

$$S = \bar{A}\bar{B}C_{in} + \bar{A}B\overline{C_{in}} + ABC_{in} + A\bar{B}\overline{C_{in}}$$

$$= C_{in}(AB + \bar{A}\bar{B}) + \overline{C_{in}}(A\bar{B} + \bar{A}B)$$

$$= \underbrace{C_{in}}_{X}\ \underbrace{\overline{(A \oplus B)}}_{\bar{Y}} + \underbrace{\overline{C_{in}}}_{\bar{X}}\ \underbrace{(A \oplus B)}_{Y}$$

$$\boxed{S = C_{in} \oplus (A \oplus B)}$$

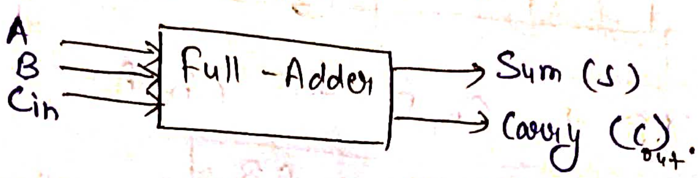
3

FULL ADDER

A Full adder is a comb. circuit that adds 2 bits & carry ~~that adds~~ and outputs a sum bit & a carry bit.

It has 3 input & 2 output.

Block Diagram



Truth table

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

S	C
0	1

K-MAP for S & Cout

for S.

Cin \ AB	00	01	11	10
0	$_0$	1_1	$_3$	1_2
1	1_4	$_5$	1_7	$_6$

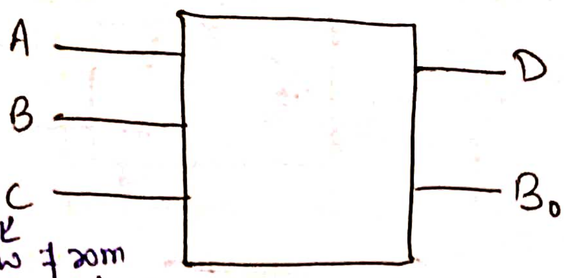
$$S = \bar{A}\bar{B}C_{in} + \bar{A}B\overline{C_{in}} + ABC_{in} + A\bar{B}\overline{C_{in}}$$

$$= C_{in}(AB + \bar{A}\bar{B}) + \overline{C_{in}}(A\bar{B} + \bar{A}B)$$

$$= \underbrace{C_{in}}_{X}\ \underbrace{\overline{(A \oplus B)}}_{\bar{Y}} + \underbrace{\overline{C_{in}}}_{\bar{X}}\ \underbrace{(A \oplus B)}_{Y}$$

$$\boxed{S = C_{in} \oplus (A \oplus B)}$$

FULL SUBTRACTOR.



(output Borrow)

Borrow from next stage.

A	B	C	D	B_0
0	- 0 = 0	- 0 = 0	0	0
0	- $\cancel{0}^2$ = 0	- 1 =	1	1
$\cancel{0}$ 2	- 1 = 1	0	1	1
0	- 1 = 1	1	0	1
1	- 0 = 1	0	1	0
1	0 = 1	- 1 = 0	0	0
1	1 = 0	- 0	0	0
1	1 = $\cancel{0}$	- 1 =	1	1

$$\left.\begin{array}{l} D = A \oplus B \oplus C \\ B_0 = BC + \bar{A}C + \bar{A}B \end{array}\right\} \text{using K-Map)}$$

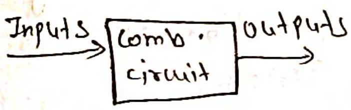
SEQUENTIAL CIRCUITS.

* The output variables at any instant of time depend on present input and also on present state i.e., on past history of the system.
* Memory unit is required to store the past history of input variables.

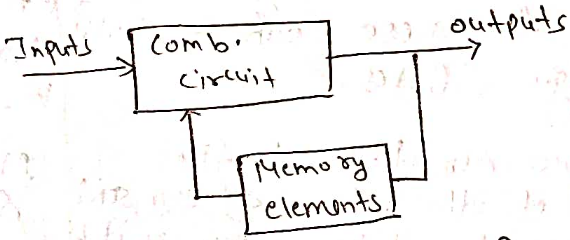
* It is slower than comb. circuit becoz delay is due to propagation delay of gates & due to memory element also.

* It is comparatively harder to design.

Ex→ Latch, Flip-Flops, Counter, Register, etc.



Block diagram of Comb. Circuit.



Block diagram of Seq. Circuit.

Seq. Circuit = C.C + M.E.

- Synchronous Sequential Circuit. (Jo clock k help se control hota h)
- Asynchronous Seq. circuit (Jo clock se control nhi hota)

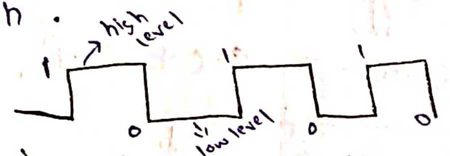
Syn. S. C	Asyn. S.C
(1). Seq. ~~Circ~~ which are controlled by a clock are called Syn. S.C.	(1). The seq. circuits which are not controlled by a clock are asy. S. C.
(2). The change in input signals can affect memory elements only on activation of clock signal.	(2). The change in input signals can affect memory elements at any instant of time.
(3). Slower than Asy. bcoz.. depends on clock.	(3). faster than syn. S.C due to absence of clock.

Syn. S.C Ex→

LATCHES

(1). Latches are building blocks of seq. C. & these can be built from logic Gates.

(2). Latches are sensitive to the duration of the pulse & can send or recieve the data when switch is on.



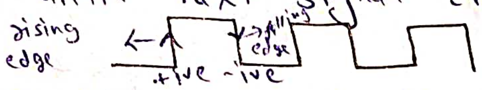
(3). It is based on the enable input.

(4). It is a level triggered, it mean, that the output of the present state & input of the next state depends on the level that is binary input 1 or 0.

Flip-flops.

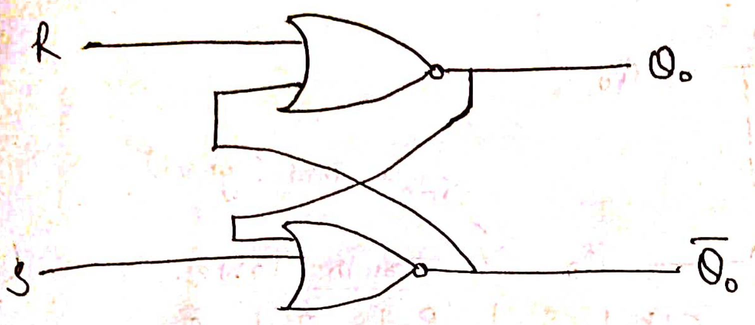
(1). Flip-flop are also building blocks of seq. C. but these can be built from latches.

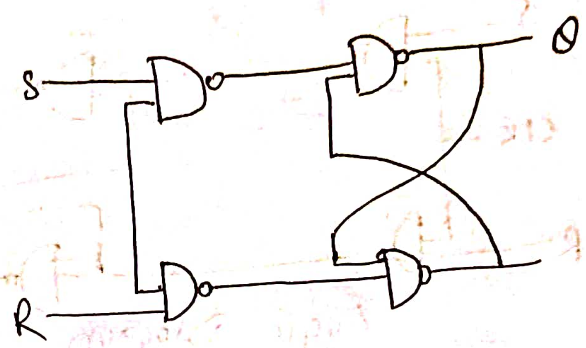
(2). Flip-flop is sensitive to a signal change. They can transfer data only at a single instant & data cannot be changed until next signal change.



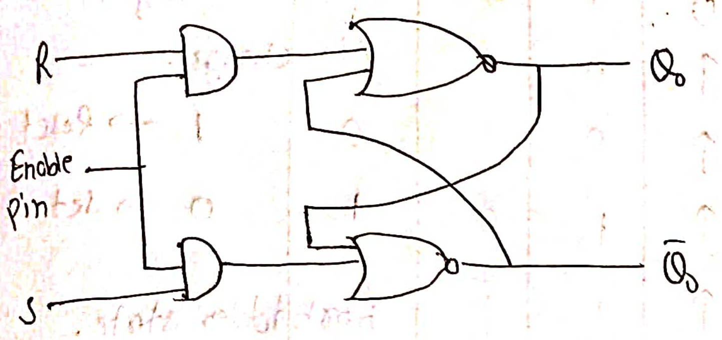
(3). It is based on clock pulses.

(4). It is an edge triggered, it means that output & next State input changes when there is a change in clock pulse whether it may be +ive or -ive clock.





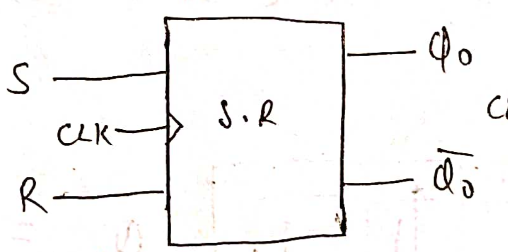
Flip - flop



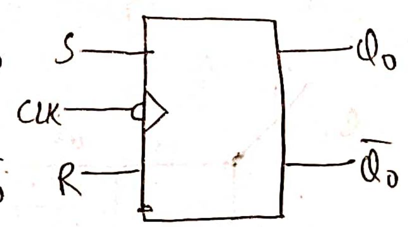
Latch

SR- flip- flop.

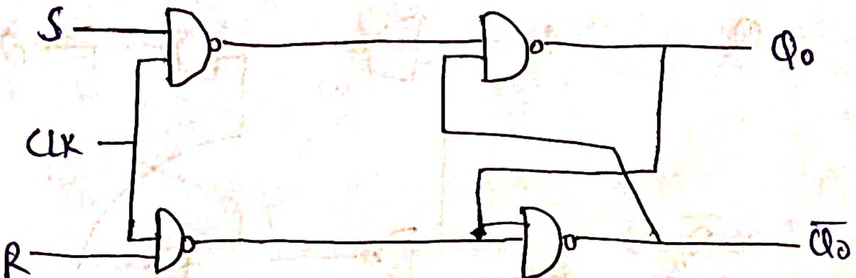
- S-R stands for Set - Reset.
- The S & R inputs of the S-R flip-flop are called synchronous control inputs because data on these inputs affect the flip-flop's output only on the triggering edge of the clock pulse.
- Without a clock pulse, the S & R inputs cannot affect the output.
- Clock pulse can be +ive or -ive.



+ive edge S-R F.F



-ive edge S-R F.F



Logic Diagram

Don't care (symbol)

Truth Table.

CLK	S	R	Q_0	$\bar{Q}_0$
0	X	X	No change	No change.
↑	0	0	No change.	
↑	0	1	0	1 → Reset
↑	1	0	1	0 → Set.
↑	1	1	Forbidden state.	

Characteristic Table.

PRESENT STATE Q_n	S	R	NEXT STATE Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X (Don't care cond^n)
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X (Don't care)

Q_n \ SR	00	01	11	10
0	(0)	1 (2)	1 (6)	1 (4)
1	(1)	X (3)	X (7)	(5)

$$Q_{n+1} = R + S\bar{Q}_n$$

characteristic eq^n.

EXCITATION TABLE.

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

S	R
0	0
0	1
0	X

S	R
1	0

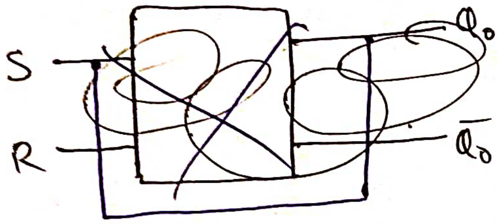
S	R
0	0
1	0
X	0

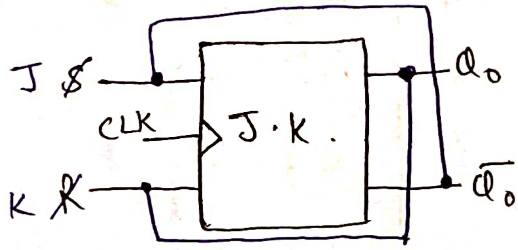
J-K FLIP-FLOP.

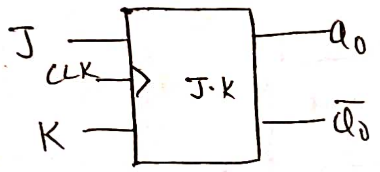
J-K stands for Jack Kilby

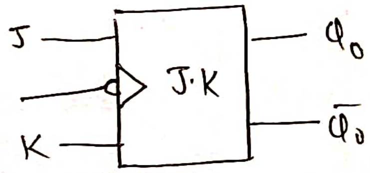
* Function of J-K is similar to S-R flip-flop, except that it has no invalid state like that of the S-R flip-flop.

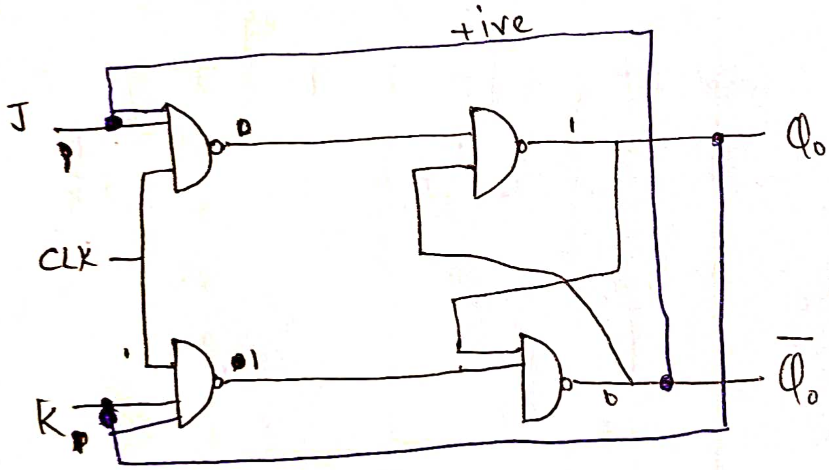
* J-K flip flop can be obtained from S-R flip-flop by connecting Q_0 to R & $\bar{Q}_0$ to S.











$\bar{Q}_0 = 0$ | $\bar{Q}_0 = 1$
$Q_0 = 1$ | $Q_0 = 1$

CLK	J	K	Q	$\bar{Q}_0$
0	X	X	No change	
↑	0	0	No change	
↑	0	1	0	1
↑	1	0	1	0
↑	1	1	Previous state complement.	

CHARACTERISTIC TABLE

PREVIOUS STATE 'Qn'	J	K	NEXT STATE 'Qn+1'
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

K \ $Q_n J$	00	01	11	10
0	(0)	1 (2)	1 (6)	1 (4)
1	(1)	1 (3)	(7)	(5)

$$\boxed{Q_{n+1} = \overline{Q_n} J + Q_n \overline{K}}$$

Characteristic eq^n

EXCITATION TABLE.

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

J	K
0	0
0	1
0	X

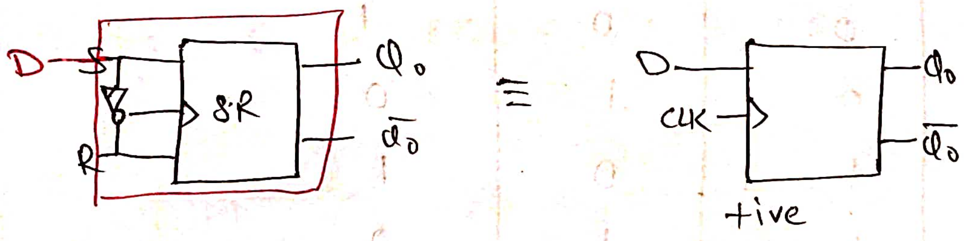
J	K
1	0
1	1
1	X

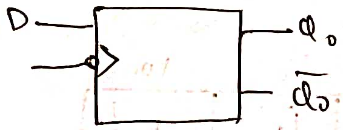
J	K
0	1
1	1
X	1

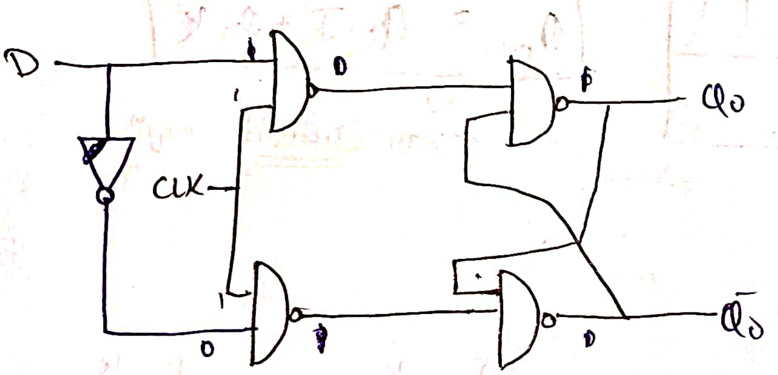
J	K
0	0
1	0
X	0

D – Flip-Flop.

→ D stands for delay. So, it is called Delay flip-flop.
→ D flip-flop has only one input terminal.
→ D flip-flop can be obtained from S-R f.f by just putting one inverter b/w S & R terminal.







CLK	D	Q_0	$\bar{Q}_0$
0	X	No change	
↑	0	0	1
↑	1	1	0

Characteristic table.

PRESENT STATE Q_n	D	NEXT STATE Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

D \ Q_n	0	1
0		
1	1	1

$$\boxed{Q_{n+1} = D}$$

EXCITATION TABLE

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

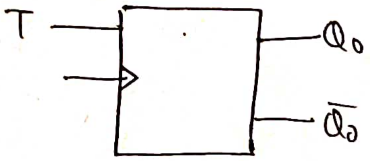
T-FLIP FLOP

→ T stands for Toggle, so it is called Toggle flip-flop.

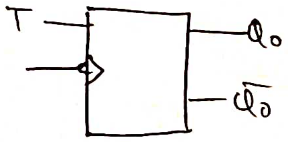
→ T.F.F. has only one input terminal.

→ T.F.F can be obtained from J.K F.F by just joining J & K terminal together.

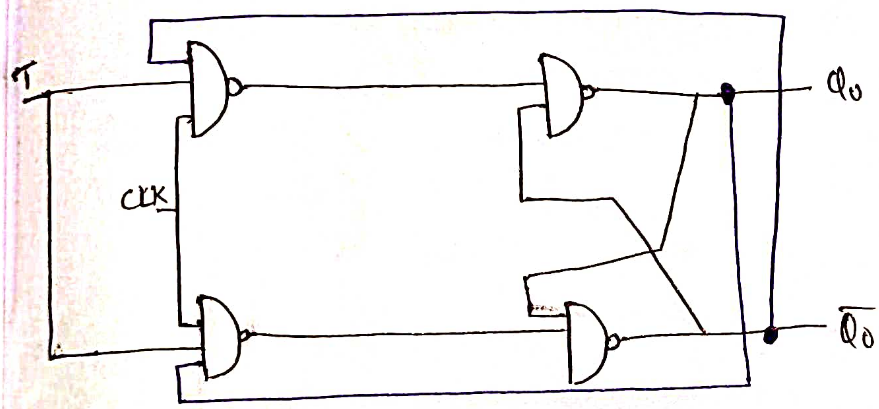
→ T F.F are not widely available commercially becoz it is easy to convert a J.K F.F to T.F.F & labelling the connection as T.



+ive edge



−ive edge



CLK	T	Q_{n+1}
0	X	Q_n
↑	0	Q_n
↑	1	$\overline{Q_n}$

PRESENT STATE Q_n	T	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

T \ Q_n	0	1
0	(0)	1 (2)
1	1 (1)	(3)

$Q_{n+1} = \overline{Q_n}T + Q_n\overline{T}$

$$\boxed{Q_{n+1} = Q_n \oplus T}$$

Exicitation Table.

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0